

INTRODUCTION

A body is at rest when it does not change its position with time and is in motion if it changes its position with time in the frame of reference of the observer.

All motion is relative. There is no meaning of rest or motion without reference to the observer.

A passenger in a moving train is at rest with respect to another passenger in the same train while both are in motion with respect to observer on the ground. Therefore nothing is at absolute rest or in absolute motion.

To describe the motion of a particle, we introduce four important quantities namely position, displacement, velocity and acceleration. In general motion of a particle in three dimension these quantities are vectors which have direction as well as magnitude. But for a particle moving in a straight line, there are only two directions, distinguished by designating one as positive and one negative.

DISTANCE AND DISPLACEMENT

Distance

The length of the actual path between initial and final positions of a particle in a given interval of time is called distance covered by the particle. It is the actual length of the path covered by the body.

Characteristics of Distance

- ❖ It is a scalar quantity
- ❖ It depends on the path
- ❖ It never reduces with time
- ❖ Distance covered by a particle is always positive or zero and can never be negative
- ❖ Dimension: $[M^0 L^1 T^0]$
- ❖ Unit: In C. G. S. centimeter (cm), In S.I. system meter (m).

Displacement

The shortest distance from the initial position to the final position of the particle is called displacement. The displacement of a particle is measured as the change in the position of the particle in a particular direction over a given time interval. It depends only on final and initial positions.

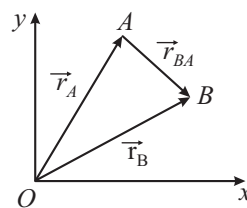
Displacement of a particle is a position vector of its final position w.r.t, initial position.

Position vector of A w.r.t. $O = \vec{OA}$

$$\Rightarrow \vec{r}_A = x_1 \hat{i} + y_1 \hat{j} + z_1 \hat{k}$$

Position Vector of B w.r.t. $O = \vec{OB}$

$$\Rightarrow \vec{r}_B = x_2 \hat{i} + y_2 \hat{j} + z_2 \hat{k}$$



$$\text{Displacement} = \vec{AB} = \vec{r}_{BA}$$

$$= (x_2 - x_1) \hat{i} + (y_2 - y_1) \hat{j} + (z_2 - z_1) \hat{k}$$

For a particle moving in a straight line, if we take x -axis along direction of motion, then displacement has only one component $S = x_2 - x_1$

Characteristics of Displacement

- ❖ It is a vector quantity.
- ❖ The displacement of a particle between any two points is equal to the shortest distance between them.
- ❖ The displacement of an object in a given time interval may be +ve, -ve or zero.
- ❖ The actual distance travelled by a particle in the given interval of time is always equal to or greater than the magnitude of the displacement and in no case, is it less than the magnitude of the displacement, i.e. $\text{Distance} \geq |\text{Displacement}|$
- ❖ Dimension: $[M^0 L^1 T^0]$
- ❖ Unit: In C. G. S. centimeter (cm), In S.I. system meter (m).

Note: Distance is always positive but displacement may be +ve, -ve or zero.

AVERAGE SPEED AND INSTANTANEOUS SPEED

Average speed is the ratio of total distance covered by a particle in a given time interval divided by the time interval.

$$v_{av.} = \frac{\Delta s}{\Delta t} \quad (\text{where } \Delta t = t_2 - t_1)$$

Instantaneous speed at any instant t is

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}$$

- ❖ The slope of the distance-time graph provides the value of instantaneous speed.
- ❖ The average speed is defined for a time interval while the instantaneous speed is defined at an instant. The word speed normally implies instantaneous speed.
- ❖ Average speed and instantaneous speed both are scalar quantities.
- ❖ For any moving object, the average speed can never be zero or negative, i.e., $v_{av} > 0$, as total distance covered is always +ve only.
- ❖ If a particle travels distances $s_1, s_2, s_3, \dots$, etc., at different speeds $v_1, v_2, v_3, \dots$ etc., respectively.

$$v_{av} = \frac{\Delta s}{\Delta t} = \frac{\sum s_i}{\sum (s_i / v_i)} \text{ If } s_1 = s_2 = \dots = s_n = s,$$

$$\text{Then } \frac{1}{v_{av}} = \frac{1}{n} \left[\frac{1}{v_1} + \frac{1}{v_2} + \dots \right] = \frac{1}{n} \sum \frac{1}{v_i}$$

Special case: If a particle moves a distance at speed v_1 and comes back to initial position with speed v_2 , then

$$v_{av} = \frac{2v_1v_2}{v_1 + v_2}$$

- ❖ If a particle travels at speeds $v_1, v_2, \dots$, etc., for time intervals $t_1, t_2, \dots$

$$v_{av} = \frac{\Delta s}{\Delta t} = \frac{v_1t_1 + v_2t_2 + \dots + v_nt_n}{t_1 + t_2 + \dots + t_n} = \frac{\sum v_it_i}{\sum t_i}$$

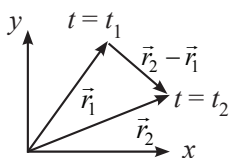
- ❖ If a particle moves for two equal intervals of time at different speeds, then $v_{av} = \frac{v_1 + v_2}{2}$.

AVERAGE VELOCITY

The average velocity is the ratio of displacement from time t_1 to t_2 and the time interval $t_2 - t_1$.

$$\vec{v}_{av} = \frac{\vec{r}_2 - \vec{r}_1}{t_2 - t_1} = \frac{\Delta \vec{r}}{\Delta t}$$

In one dimension, $v_{av} = \frac{x_2 - x_1}{t_2 - t_1}$



INSTANTANEOUS VELOCITY

The instantaneous velocity at any instant t is

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k}$$

The magnitude of instantaneous velocity is always equal to the instantaneous speed.

Instantaneous speed

$$= |\vec{v}| = \lim_{\Delta t \rightarrow 0} \frac{|\Delta \vec{r}|}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}$$

Instantaneous velocity is called simply velocity.

In one dimension, we can simply write velocity as

$$v = v_x = \frac{dx}{dt}$$

If an object moves along a straight line without changing its direction, then the magnitude of the average velocity is equal to the average speed, otherwise magnitude of average velocity < average speed.

- ❖ Velocity can be +ve or -ve as it is a vector but speed can never be negative as it is the magnitude of velocity.
- ❖ If a body is moving with a constant velocity, then the average velocity and instantaneous velocity are equal.
- ❖ The velocity of a body is uniform, if both magnitude and direction do not change.
- ❖ If a body moves with non-uniform velocity, then magnitude of velocity may change or direction of velocity may change or both.
- ❖ A body can have non-zero speed and zero average velocity when a body completes one revolution around a circle, the average velocity is zero since the displacement is zero. But the average speed is not zero since the distance travelled $\neq 0$.
- ❖ If a body is moving with constant speed then its velocity may or may not be constant. In case of uniform circular motion though speed remains constant but velocity changes instant to instant because of change in direction.

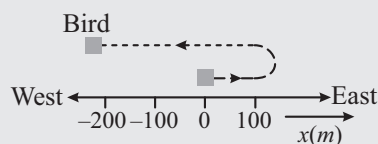


Train Your Brain

Example 1: A bird flies towards east at 10 m/s for 100 m. It then turns around and flies at 20 m/s for 15s. Find

- (a) Its average speed (b) Its average velocity

Sol. (a) Let us take the x axis to point eastwards. A sketch of the path is shown in the figure. To find the required quantities, we need the total time interval. The first of the journey took.



$\Delta t_1 = (100 \text{ m}) / (10 \text{ m/s}) = 10\text{s}$, and we are given $t_2 = 15\text{s}$ for the second part. Hence the total time interval is $\Delta t = \Delta t_1 + \Delta t_2 = 25 \text{ s}$

The bird flies 100 m east and then $(20 \text{ m/s}) \times (15\text{s}) = 300 \text{ m west}$.

$$(a) \text{ Average speed} = \frac{\text{Distance}}{\Delta t}$$

$$= \frac{100 \text{ m} + 300 \text{ m}}{25 \text{ s}} = 16 \text{ m/s}$$

(b) The net displacement is
 $\Delta x = -200 \text{ m}$

$$\text{So, } v_{av} = \frac{\Delta x}{\Delta t} = \frac{-200 \text{ m}}{25 \text{ s}} = -8 \text{ m/s}$$

The negative sign means that v_{av} is directed toward the west.

Example 2: A particle moves with speed v_1 along a particular direction. After some time it turns back and reaches the starting point again travelling with speed v_2 . Find (for the whole journey).

(a) Average velocity (b) Average speed

Sol. (a) Since the particle reaches the starting point again, its displacement is zero.

$$\therefore \text{Average velocity} = \frac{\text{Net displacement}}{\text{total time}} = 0$$

(b) Let it travelled distance x while moving away as well as while moving towards the starter point.

$$\text{Time taken to go away is } t_1 = \frac{x}{v_1}$$

$$\text{Time taken while return journey } t_2 = \frac{x}{v_2}$$

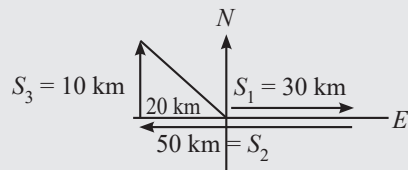
$$\therefore \text{Average speed} = \frac{x+x}{t_1+t_2} = \frac{2x}{\frac{x}{v_1} + \frac{x}{v_2}}$$

$$v_{av} = \frac{2v_1v_2}{v_1+v_2}$$

i.e., harmonic mean of individual speeds.

Example 3: A person goes 30 km East, then he walks 50 km west and then he goes 10 km N. Find Average speed and average velocity for the whole journey in 15 hrs.

Sol.



$$\text{Average speed} = \frac{\text{Total Distance}}{\text{Total Time}} = \frac{30 + 50 + 10}{15}$$

$$= 6 \text{ km/hr}$$

$$\text{Average velocity} = \frac{\text{Total Displacement}}{\text{Total Time}}$$

$$= \frac{\sqrt{(20^2 + 10^2)}}{15} = \frac{10\sqrt{5}}{15}$$

$$\text{Average velocity} = \frac{2\sqrt{5}}{3} \text{ km/hr}$$



Concept Application

1. A car moves 30 km with 20 km/hr and then 30 km with 30 km/hr. Find average speed in Average velocity for the whole journey in the same straight line.

ACCELERATION

The rate of change of velocity of an object with time is called acceleration of the object.

Let v and v' be the velocity of the object at time t and t' respectively, then acceleration of the body is given by

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{dv_x}{dt}\hat{i} + \frac{dv_y}{dt}\hat{j} + \frac{dv_z}{dt}\hat{k}$$

For one dimensional motion,

$$a = a_x = \frac{dv_x}{dt} = \frac{d\vec{v}}{dt}$$

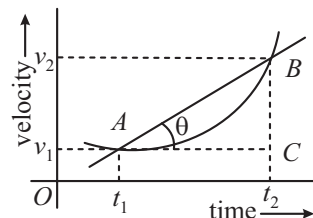
- ❖ Acceleration is a vector quantity.
- ❖ It is positive if the velocity is increasing and is negative if the velocity is decreasing.
- ❖ The negative acceleration is also called retardation or deceleration.
- ❖ **Unit:** In S.I. System m/s^2
In C.G.S. System cm/s^2
- ❖ **Dimension :** $[M^0L^1T^{-2}]$

Average Acceleration

When an object is moving with a variable acceleration in a straight line, then the average acceleration of the object for the given motion is defined as the ratio of the total change in velocity of the object during motion to the total time taken i.e.,

$$\text{Average Acceleration} = \frac{\text{total change in velocity}}{\text{total time taken}}$$

Suppose the velocity of a particle is v_1 at time t_1 and v_2 at time t_2 .



Then, Change in velocity $= v_2 - v_1 = \Delta v$

Elapsed time in changing the velocity $= t_2 - t_1 = \Delta t$

$$\text{Thus, } a_{av} = \frac{v_2 - v_1}{t_2 - t_1} = \frac{\Delta v}{\Delta t}$$

$\Rightarrow a_{av} = \frac{BC}{AC} = \tan \theta$ = the slope of chord of $v - t$ graph is average acceleration.